# UNIFY

Universal AI Implementation Engine

# Stakeholder Project Summary & Implementation Plan

A vendor-first AI platform designed to automate repeatable ERP customer implementation work—from discovery and data migration to configuration, testing and go-live readiness.

| PRODUCT DIRECTION | FIRST PROTOTYPE | INITIAL HORIZON |
|---|---|---|
| Commercial B2B SaaS | ERP SaaS implementation | 6-month MVP plan |

## Stakeholder decision in one sentence

**Build a reusable implementation execution platform for ERP software vendors, starting with a controlled workflow that maps, validates and migrates customer master data into an ERP sandbox.**

One governed workflow, with human approval at material decision points

## Purpose of this document

Provide stakeholders with a concise view of the product opportunity, proposed scope, end-to-end operating flow, solution architecture, delivery plan, pilot measures, key risks and decisions required to authorize the next stage.

Status: proposed product and delivery baseline for discussion. Product name, market claims, costs, dates and performance targets remain subject to validation.

# 1. Executive summary

UNIFY is proposed as an AI-powered implementation execution platform sold to B2B software vendors. The vendor registers its ERP product, connects a supported sandbox, and uses UNIFY to manage and automate repeatable customer implementation activities.

| Dimension | Proposed direction |
|---|---|
| Primary buyer | ERP / SaaS product vendors with recurring customer implementations. |
| Users | Vendor admins, implementation managers, consultants, migration specialists, QA and customer admins. |
| Initial use case | Customer onboarding, requirements, implementation planning, master data validation/mapping and sandbox migration. |
| Product model | Multi-tenant SaaS with a product-independent core and product-specific connectors / playbooks. |
| Execution model | AI proposes and coordinates; deterministic services execute; people approve material decisions. |
| MVP horizon | Indicative six months, subject to team capacity, API access and pilot partner readiness. |

## Why start with ERP?

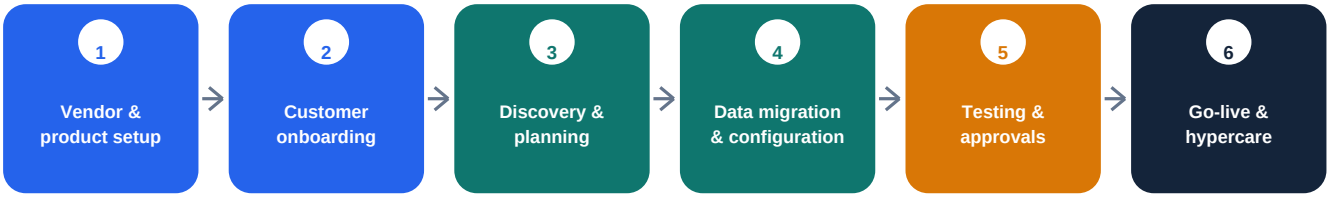
ERP implementation is a demanding proving ground: it combines structured master data, business rules, configuration, integrations, testing and reconciliation. Demonstrating reliable execution in one ERP sandbox will test the core engine before expanding to WMS, logistics, manufacturing or other configurable B2B software.

## What UNIFY is—and is not

| UNIFY IS | UNIFY IS NOT |
|---|---|
| An implementation execution and orchestration layer that connects to an ERP product and automates approved implementation work. | A replacement ERP, a generic chatbot, or merely a project tracker that records tasks without executing implementation activities. |

# 2. Overall product flow

The lifecycle below shows how a vendor uses UNIFY to take a new ERP customer from setup through go-live readiness. The workflow is governed by approvals and evidence at key stages.




One governed workflow, with human approval at material decision points

## Lifecycle stages and outcomes

| Stage | What happens | Primary output |
|---|---|---|
| 1. Vendor & product setup | Register vendor, ERP product, modules, entities, connector and sandbox. | Configured product profile |
| 2. Customer onboarding | Capture customer profile, scope, modules, users, documents and target dates. | Customer implementation record |
| 3. Discovery & planning | Extract requirements, identify fit-gap, clarify unknowns and generate plan/dependencies. | Approved requirements and plan |
| 4. Migration & configuration | Profile files, map fields, validate, transform and execute approved sandbox actions. | Migrated data and configuration evidence |
| 5. Testing & approvals | Run agreed scenarios, reconcile outcomes, resolve exceptions and obtain sign-offs. | Test evidence and approval record |
| 6. Go-live & hypercare | Assess readiness, obtain authorized cutover approval and monitor early issues. | Readiness decision and hypercare record |

## MVP boundary

The first release should prove one real implementation path, not attempt universal compatibility immediately. Use a mock ERP API for early development, then connect one real vendor sandbox. Production migration and autonomous production go-live are explicitly outside the initial MVP.

# 3. First flagship workflow: ERP master data migration

The first demonstrable value should be a controlled customer-master migration. A customer uploads a spreadsheet; UNIFY profiles it, recommends mappings, applies deterministic rules, requests approval, migrates to the ERP sandbox and reconciles the outcome.

*Exceptions return to data correction; production execution is outside the initial MVP.*

## Illustrative migration example

| Source column | ERP target field | Treatment |
|---|---|---|
| Cust_Code | customer_code | Candidate exact mapping; uniqueness check. |
| Cust_Name | customer_name | Candidate exact mapping; mandatory-field check. |
| GSTIN_No | tax_registration_no | Format and jurisdiction validation; review if ambiguous. |
| Opening_Due | opening_balance | Confirm sign, currency, accounting date and business meaning. |

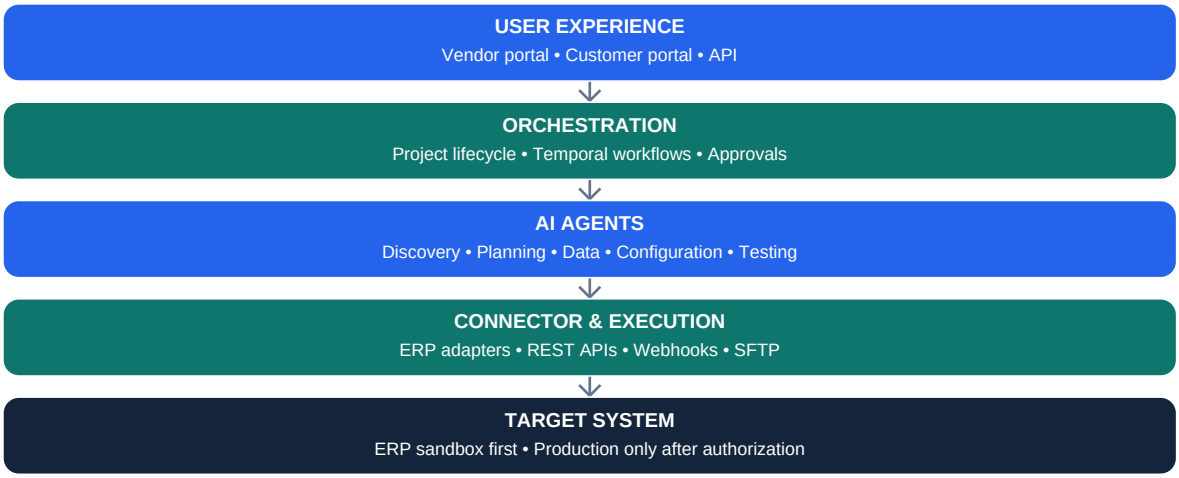
## Mandatory controls

Preserve the original source file; version mappings and transformation rules; validate required fields and references; execute in batches; maintain record-level traceability; make retries idempotent; reconcile counts and applicable balances; and retain an exception report. No production write is permitted without explicit authorization.

Illustrative output: a migration report with source count, accepted count, rejected count, created/updated target count, reconciliation result and downloadable exceptions.

# 4. Solution architecture

The recommended starting point is a modular monolith with independently running background workers. This keeps the first release simpler to build and operate while preserving boundaries that can later be split into services.




## Recommended technology choices

| Layer | Initial recommendation | Reason |
|---|---|---|
| Web application | React / Next.js | Vendor and customer portals. |
| Backend/API | Python / FastAPI | Typed APIs and Python data ecosystem. |
| Primary database | PostgreSQL + pgvector | Transactional data and initial semantic retrieval. |
| Workflow engine | Temporal | Durable workflows, retries, timers and recovery. |
| Agent orchestration | LangGraph | Specialized agent routing and human-in-the-loop steps. |
| AI access | Model gateway; OpenAI API initially | Model abstraction, usage control and observability. |
| Files | S3-compatible object storage | Original uploads, reports and generated artifacts. |
| Monitoring | OpenTelemetry | Traces, metrics and execution diagnostics. |

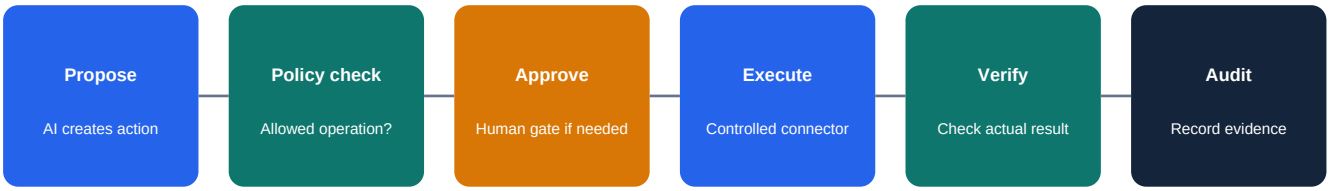
**Architecture rule:** AI reasons and proposes; deterministic services validate and execute; product adapters expose only documented capabilities; workflow orchestration records state and recovery; humans approve material actions.

# 5. AI agents and governance

Use specialized agents coordinated by an implementation orchestrator. The durable project workflow should be controlled by the workflow engine; agents should not independently decide to bypass project states or permissions.

| Agent | Responsibility | MVP output |
|---|---|---|
| Discovery | Extract and classify requirements; identify gaps and questions. | Reviewed requirements baseline |
| Planning | Build phases, tasks, dependencies and milestone suggestions. | Approved implementation plan |
| Data migration | Profile, map, transform, validate and reconcile records. | Sandbox migration report |
| Configuration | Propose supported configuration actions and verify results. | Configuration evidence |
| Integration | Configure and test supported external connections. | Integration test report |
| Testing | Generate and execute requirement-linked scenarios. | Test execution evidence |
| Go-live | Evaluate mandatory readiness conditions and blockers. | Readiness recommendation |
| Hypercare | Classify issues and suggest approved knowledge-based resolutions. | Issue triage recommendations |

## Governed execution pattern

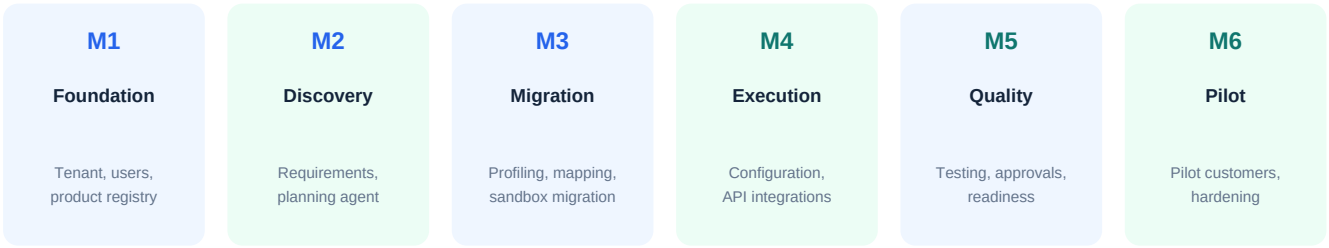



## Approval policy

| Activity | Recommended control |
|---|---|
| Document extraction / data profiling | Automatic processing; human review of uncertain findings. |
| Requirements and plan | Consultant / implementation manager approval before baseline. |
| Field mapping and transformation | Specialist approval for ambiguous or material mappings. |
| Sandbox migration / configuration | Authorized project approval and logged execution. |
| Production migration / go-live | Explicit authorized approval; not autonomous in MVP. |

# 6. Six-month project plan

Indicative 24-week delivery plan, assuming a small experienced product and engineering team, timely access to an ERP sandbox/API, and an engaged pilot vendor. Dates should be baselined after discovery and resource confirmation.




| Period | Focus | Key work | Exit deliverable |
|---|---|---|---|
| Month 1 | Foundation | Architecture, tenancy, identity, roles, product registry, connector contract and mock ERP. | Vendor can register a product and test a sandbox connector. |
| Month 2 | Discovery & planning | Customer onboarding, document ingestion, discovery agent, requirements review and plan generation. | Approved implementation blueprint. |
| Month 3 | Data migration | Excel/CSV processing, profiling, field mapping, validation, transformation and sandbox migration. | Traceable migration and reconciliation report. |
| Month 4 | Configuration & integration | Supported ERP configuration actions, REST integrations, webhooks and execution logging. | Controlled configuration and integration test. |
| Month 5 | Testing & readiness | Requirement-linked tests, approvals, exception handling, audit and readiness report. | End-to-end controlled implementation workflow. |
| Month 6 | Pilot & hardening | Connect first real ERP sandbox, run pilot projects, measure outcomes, improve security and usability. | Pilot-ready MVP and evidence-based business case. |

## Delivery gates

**Gate 1 – End of Month 1:** tenant isolation, product registry and mock connector demonstrated.
**Gate 2 – End of Month 2:** requirements and plan reviewed by a real implementation practitioner.
**Gate 3 – End of Month 3:** sandbox migration and reconciliation demonstrated.
**Gate 4 – End of Month 5:** end-to-end controlled workflow passes acceptance tests.
**Gate 5 – End of Month 6:** pilot evidence supports a go / revise / stop decision.

# 7. Team, budget and pilot

## Indicative lean team

| Role | Indicative allocation | Accountability |
|---|---|---|
| Product / implementation architect | 1 full-time or founder-led | Product scope, workflows, acceptance criteria, pilot and domain rules. |
| AI / backend engineers | 2 | Agent runtime, APIs, workflow integration and data services. |
| Full-stack engineer | 1 | Vendor portal, customer portal and dashboards. |
| Integration engineer | 1 | ERP adapter, API connectivity and sandbox execution. |
| QA / automation engineer | 1 | Test automation, migration validation and release quality. |
| UX designer | Part-time | User flows and interaction design. |

## Indicative six-month budget

| Cost category | Indicative range | Notes |
|---|---|---|
| Engineering team | ■30–45 lakh | Largest cost; depends on hiring mix and founder contribution. |
| Cloud and AI infrastructure | ■2–5 lakh | Early-stage environments, model usage and storage. |
| Security and testing | ■2–4 lakh | Security review, testing tools and remediation allowance. |
| Design, legal and miscellaneous | ■2–4 lakh | UX support, contracts and operating expenses. |
| Indicative total | ■36–58 lakh | Planning estimate; excludes major certification and significant sales costs. |

## Pilot design

Secure one ERP vendor design partner with sandbox access and API documentation. Select two or three customer implementation cases, beginning with customer, vendor and item master migration. Establish a baseline for manual hours, elapsed time, exceptions and rework before using UNIFY; compare like-for-like outcomes after the pilot.

## Pilot success measures

| Measure | Evidence to collect |
|---|---|
| Effort reduction | Consultant hours per activity before vs. after UNIFY. |
| Migration quality | Source-to-target traceability, rejected records, duplicate prevention and reconciliation. |
| Cycle time | Elapsed time from file receipt to approved sandbox migration. |
| Intervention rate | Number and type of human corrections or approvals per run. |
| Reliability | Execution failures, retry outcomes, recovery time and audit completeness. |

# 8. Risks, dependencies and decisions

## Key risks and mitigations

| Risk / dependency | Why it matters | Mitigation / owner |
|---|---|---|
| ERP API access is limited | Automation cannot reliably configure or migrate through unsupported interfaces. | Secure a design partner and API contract early; start with mock connector. Product / integration lead. |
| Inconsistent customer data | AI mapping alone cannot resolve ambiguous business meaning. | Deterministic rules, source traceability, exception queue and specialist approvals. Data lead. |
| Unsafe agent execution | Incorrect writes can affect financial or operational records. | Allow-listed tools, least privilege, sandbox-first execution, approvals and audit. Engineering / security. |
| Tenant data leakage | UNIFY will process customer and financial data across vendors. | Tenant-scoped authorization, RLS, security tests, secrets manager and isolation review. Platform lead. |
| Scope expansion | Universal product support can delay proof of value. | One ERP connector and limited entities for MVP; add adapters only after pilot evidence. Product lead. |
| Unproven willingness to pay | Technical success does not prove commercial demand. | Interview vendors, quantify implementation cost and obtain pilot commitments before scaling. Founder / sales. |

## Stakeholder decisions requested

**1.** Approve the vendor-first ERP SaaS positioning and the proposed MVP boundary.

**2.** Nominate a product / implementation owner and a technical lead.

**3.** Authorize discovery with prospective ERP vendor design partners.

**4.** Secure access to one ERP sandbox, API documentation and representative test data.

**5.** Agree the six-month planning horizon and validate the ■36–58 lakh indicative budget.

**6.** Approve a gated approach: foundation → migration proof → end-to-end MVP → pilot go / no-go.

## Immediate next steps (first 2 weeks)

• Confirm stakeholder sponsor, product owner and engineering lead.
• Interview 5–8 ERP vendors to validate pain points, implementation volumes and buying criteria.
• Select one design partner and obtain sandbox/API access.
• Agree the first three migration entities and business validation rules.
• Convert this summary into a detailed PRD, architecture decision record and sprint backlog.
• Baseline the budget, team availability and delivery dates.

## Proposed stakeholder discussion outcome

**A decision to proceed with a time-boxed discovery and technical proof of concept—not yet an unconditional commitment to the full six-month build. The proof should establish ERP connectivity, reliable data mapping, controlled sandbox migration, reconciliation and customer/vendor willingness to pilot.**

Prepared as a discussion draft. All estimates, scope boundaries and delivery assumptions should be validated with stakeholders before being treated as approved commitments.